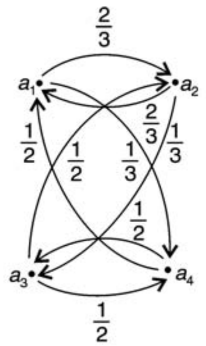

Fig. 31.6

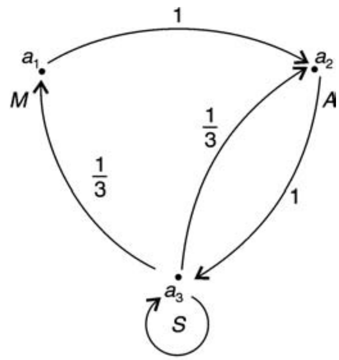

Fig. 31.7

Markov chain

Example 8: Every year, a man trades his car for a new car. If he has a Maruti, he trades it for an Ambassador. If he has an Ambassador, he trades it for a Santro. However, if he has a Santro, he is just as likely to trade it for a new Santro as to trade it for a Maruti or an Ambassador. In 2000 he bought his first car, which was a Santro.

(i) Find the probability that he has
  (a) 2002 Santro
  (b) 2002 Maruti
  (c) 2003 Ambassador
  (d) 2003 Santro

(ii) In the long run, how often will he have a Santro.

Solution: **(i)** Define 3 states a_1, a_2, a_3 as follows a_1: state of having Maruti car, a_2: having Ambassador, a_3: having Santro. Then the transition matrix is

$$P = \begin{array}{c} \\ a_1 \\ a_2 \\ a_3 \end{array} \begin{array}{c} \begin{array}{ccc} a_1 & a_2 & a_3 \end{array} \\ \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix} \end{array}$$

2000: Initial state $= P^{(0)} = (0, 0, 1)$ since he has Santro car in 2000 (his first purchase).

(a) To reach 2002 year, (2-steps later) compute the 2-step transition matrix P^2

$$P^2 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{9} & \frac{4}{9} & \frac{4}{9} \end{pmatrix}$$

Then

$$p^{(2)} = p^{(0)} P^2 = (0 \quad 0 \quad 1) \begin{pmatrix} 0 & 0 & 1 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{9} & \frac{4}{9} & \frac{4}{9} \end{pmatrix}$$

$$p^{(2)} = \begin{pmatrix} \frac{1}{9} & \frac{4}{9} & \frac{4}{9} \end{pmatrix}$$

The probability that he has a Santro in the year 2002 is $p_3^{(2)} = \frac{4}{9}$.

(b) Probability that he has a Maruti in 2002 is $p_1^{(2)} = \frac{1}{9}$.

(c) To reach 2003: 3 steps later

$$p^{(3)} = p^{(2)} P = \begin{pmatrix} \frac{1}{9} & \frac{4}{9} & \frac{4}{9} \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix}$$

(or equivalently $p^{(3)} = p^{(0)} P^3$).

$$p^{(3)} = \begin{pmatrix} \frac{4}{27} & \frac{7}{27} & \frac{16}{27} \end{pmatrix}$$

Probability that he has an Ambassador in 2003 is $p_2^{(3)} = \frac{7}{27}$.

(d) Probability that has a Santro in 2003 is

$$p_3^{(3)} = \frac{16}{27}.$$

(ii) To discover what happens in the long run, we must find a fixed probability vector t of P. Let $t = (x, y, 1-x-y)$.
Then $tP = t$

$$(x \quad y \quad 1-x-y)\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix} = (x \quad y \quad 1-x-y)$$

or $\quad 1-x-y = 3x$

$$3x + (1-x-y) = 3y$$

$$3y + (1-x-y) = 3(1-x-y)$$

Solving $\; y = \frac{1}{3}, x = \frac{1}{6}, z = \frac{3}{6} = \frac{1}{2}$

Thus $\quad t = \left(\frac{1}{6}, \frac{1}{3}, \frac{1}{2}\right)$

In the long run, he has a Santro 50% $\left(\frac{1}{2}\right)$ of the time.

Example 9: Suppose an urn A contains 2 white marbles and urn B contains 4 red marbles. At each step of the process, a marble is selected at random from each urn and the two marbles selected are interchanged. Let X_n denote the number of red marbles in urn A after n interchanges.

(i) Find the transition matrix P.

(ii) What is the probability that there are 2 red marbles in urn A after 3 steps.

(iii) In the long run, what is the probability that there are 2 red marbles in urn A.

(iv) What is the stationary distribution of the system.

Solution: There are three states a_0, a_1 and a_2 as shown in Fig. 31.8 below:

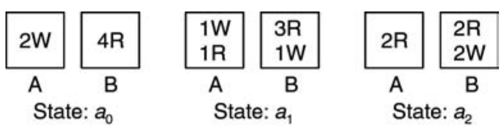

Fig. 31.8

(i) Transition matrix
If the system is in the state a_0, then a white marble from A and a red from B must be selected, so that the system will now move to state a_1. Accordingly the first row of the transition matrix (T.M.) is $(0, 1, 0)$.
Now suppose the system is in a_1. It can move to state a_0, iff red from A and white from B with probability $\frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$. Thus $p_{10} = \frac{1}{8}$. The system can move from a_1 to a_2, iff white from A and red from B with probability $\frac{1}{2} \cdot \frac{3}{4} = \frac{3}{8}$ i.e., $p_{12} = \frac{3}{8}$, probability that system will remain in a_1 itself is $1 - \frac{1}{8} - \frac{3}{8} = \frac{1}{2}$. (Note: white from A and white from B with probability $\frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$ or red from A and red from B with probability $\frac{1}{2} \cdot \frac{3}{4} = \frac{3}{8}$. Thus the probability that system will remain in state a_1 itself is $\frac{1}{8} + \frac{3}{8} = \frac{1}{2}$). Thus the 2nd row of T.M. is $\left(\frac{1}{8}, \frac{1}{2}, \frac{3}{8}\right)$.

Finally, suppose the system is in state a_2. Note that the system can never move from state a_2 to a_0. However, it may remain in a_2 itself, if a red from A and red from B is chosen. In this case the probability is $\frac{1}{1} \cdot \frac{2}{4} = \frac{1}{2}$. Lastly, if a red from A and white from B is chosen, then system moves from a_2 to a_1 with probability $\frac{2}{4} = \frac{1}{2}$. Thus third row of the T.M. is $\left(0, \frac{1}{2}, \frac{1}{2}\right)$. The Transition Matrix and transition diagram are shown in Fig. 31.9

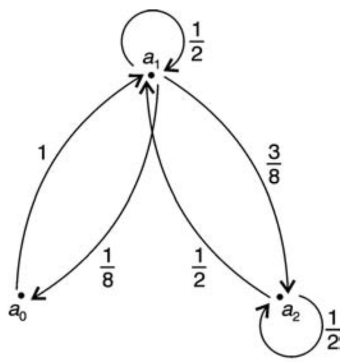

Fig. 31.9

$$\begin{array}{c c} & \begin{array}{ccc} a_0 & a_1 & a_2 \end{array} \\ \begin{array}{c} a_0 \\ a_1 \\ a_2 \end{array} & \begin{pmatrix} 0 & 1 & 0 \\ \frac{1}{8} & \frac{1}{2} & \frac{3}{8} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix} \end{array}$$

(ii) The system starts in state a_0, so that $p^{(0)} = (1, 0, 0)$ is the initial state. Now

$$p^{(1)} = p^{(0)}P = (1\ 0\ 0)\begin{pmatrix} 0 & 1 & 0 \\ \frac{1}{8} & \frac{1}{2} & \frac{3}{8} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix} = (0\ 1\ 0)$$

$$p^{(2)} = p^{(1)}P = (0\ 1\ 0)\begin{pmatrix} 0 & 1 & 0 \\ \frac{1}{8} & \frac{1}{2} & \frac{3}{8} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix} = \left(\frac{1}{8}\ \frac{1}{2}\ \frac{3}{8}\right)$$

$$p^{(3)} = p^{(2)}P = \left(\frac{1}{8}\ \frac{1}{2}\ \frac{3}{8}\right)\begin{pmatrix} 0 & 1 & 0 \\ \frac{1}{8} & \frac{1}{2} & \frac{3}{8} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix} = \left(\frac{1}{16}\ \frac{9}{16}\ \frac{6}{16}\right)$$

Probability that there are two red in A i.e., in state a_2 *after* three steps is $\frac{6}{16} = \frac{3}{8}$.

(iii) To study the system in the long run, we should find a unique fixed probability vector t of the transition matrix P. Let t be (x, y, z) or $(x, y, 1 - x - y)$.

Then $$tT = t$$

$$(x\ y\ z)\begin{pmatrix} 0 & 1 & 0 \\ \frac{1}{8} & \frac{1}{2} & \frac{3}{8} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix} = (x\ y\ z)$$

Solving
$$\frac{1}{8}y = x \quad \text{or} \quad y = 8x$$
$$x + \frac{1}{2}y + \frac{1}{2}z = y \quad \text{or} \quad 2x - y + z = 0$$
$$\frac{3}{8}y + \frac{1}{2}z = z \quad \text{or} \quad 3y = 4z$$

Now $3y = 4z = 4(1 - x - y)$

$$7y = 4 - 4x \quad \text{or} \quad 56x + 4x = 4$$

$$\therefore \quad x = \frac{4}{60},\ y = \frac{8}{15},\ z = \frac{6}{15}$$

Therefore, the fixed vector

$$t = \left(\frac{1}{15}\ \ \frac{8}{15}\ \ \frac{6}{15}\right)$$

Hence the system in the long run stays in the state a_2, 40% of the time $\left(\frac{6}{15} = \frac{2}{5}\right)$ (i.e., there will be 2 red in A, 40% of the time).

(iv) The fixed unique probability vector $t = \left(\frac{1}{15}, \frac{8}{15}, \frac{6}{15}\right)$ is the stationary distribution, since P^n approaches t, in the long run.

Markov chain with absorbing states:

Example 10: A player has Rs. 300. At each play of a game, he losses Rs. 100 with probability $\frac{3}{4}$ but wins Rs. 200 with probability $\frac{1}{4}$. He stops playing if he has lost his Rs. 300 or he has won at least Rs. 300.

(a) Determine the transition probability matrix of the Markov chain.

(b) Find the probability that there are at least 4 plays to the game.

Solution: (a) This is random walk with absorbing barriers at states 0 and 6. The transition probability matrix P is

$$P = \begin{array}{c} \\ a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \\ a_6 \end{array}\begin{array}{c} \begin{array}{ccccccc} a_0 & a_1 & a_2 & a_3 & a_4 & a_5 & a_6 \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3}{4} & 0 & 0 & \frac{1}{4} & 0 & 0 & 0 \\ 0 & \frac{3}{4} & 0 & 0 & \frac{1}{4} & 0 & 0 \\ 0 & 0 & \frac{3}{4} & 0 & 0 & \frac{1}{4} & 0 \\ 0 & 0 & 0 & \frac{3}{4} & 0 & 0 & \frac{1}{4} \\ 0 & 0 & 0 & 0 & \frac{3}{4} & 0 & \frac{1}{4} \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \end{array}$$

Explanation for the probability matrix P: There are seven states $a_0, a_1, a_2, a_3, a_4, a_5, a_6$ where a_i indicates the state that he has Rs. i hundreds (i.e., a_3 indicate he has Rs. 300, a_5 he has 500 etc.).

First row: If he is in state a_0 he has zero money (lost his initial amount of Rs. 300) and therefore he stops the game. Then he remains in that state forever. He does not play again. Thus the state a_0 is an absorbing state, no money (and he does not transit from this state to any other state).

Last row: Similarly if he has Rs. 600 (i.e., his original Rs. 300 and winning amount of Rs. 300) he stops the

game and remains in the state (he does not play). Thus the state a_6 is also an absorbing state, all money.
Second row: Suppose he has Rs. 100 i.e., he is in state a_1. Then the probability that he will lose Rs. 100 is $\frac{3}{4}$ and therefore transfers to a_0 state (no money). But he can not transfer to states a_2, a_3, a_5, a_6. But by winning Rs. 200 with probability $\frac{1}{4}$ he can have Rs. 300 (Rs. 100 original + Rs. 200 winning). Thus the probability of going from state a_1 to a_3 is $\frac{1}{4}$, and from a_1 to a_0 is $\frac{3}{4}$. Thus the probability vector (second row) is $\left(\frac{3}{4}\ 0\ 0\ \frac{1}{4}\ 0\ 0\ 0\right)$.

Third row: Starting with Rs. 200 (a_2 state) he can lose Rs. 100 with probability $\frac{3}{4}$ thereby go to state a_1 or win Rs. 200 with probability $\frac{1}{4}$, thereby go to state a_4. Thus third row $\left(0\ \frac{3}{4}\ 0\ 0\ \frac{1}{4}\ 0\ 0\right)$.
Similarly, other rows of P are obtained.
(b) The initial probability distribution is

$$p^{(0)} = (0\ \ 0\ \ 0\ \ 1\ \ 0\ \ 0\ \ 0)$$

because he has started the game with an initial amount of Rs. 300 and is therefore in state a_3. To find the probability that the game has at least 4 plays, we compute $p^{(4)}$, which gives the probability distribution of the system after 4 steps (i.e., 4 games). Now

$$p^{(1)} = p^{(0)}P =$$

$$= (0\ \ 0\ \ 0\ \ 1\ \ 0\ \ 0\ \ 0)\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3}{4} & 0 & 0 & \frac{1}{4} & 0 & 0 & 0 \\ 0 & \frac{3}{4} & 0 & 0 & \frac{1}{4} & 0 & 0 \\ 0 & 0 & \frac{3}{4} & 0 & 0 & \frac{1}{4} & 0 \\ 0 & 0 & 0 & \frac{3}{4} & 0 & 0 & \frac{1}{4} \\ 0 & 0 & 0 & 0 & \frac{3}{4} & 0 & \frac{1}{4} \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$p^{(1)} = (0, 0, \frac{3}{4}, 0, 0, \frac{1}{4}, 0)$$

$$p^{(2)} = p^{(1)}P = \left(0, \frac{9}{16}, 0, 0, \frac{6}{16}, 0, \frac{1}{16}\right)$$

$$p^{(3)} = p^{(2)}P = \left(\frac{27}{64}, 0, 0, \frac{27}{64}, 0, 0, \frac{10}{64}\right)$$

$$p^{(4)} = p^{(3)}P = \left(\frac{27}{64}, 0, \frac{81}{256}, 0, 0, \frac{27}{256}, \frac{10}{64}\right)$$

He plays 4 or more games, if after 4 steps he is not in any one of the absorbing states a_0 or a_6. Thus the probability that there at least 4 plays in the game is

$$0 + \frac{81}{256} + 0 + 0 + \frac{27}{256} = \frac{108}{256} = \frac{27}{64}.$$

Exercise

1. Which vectors are probability vectors (i) $\left(\frac{1}{2}, \frac{1}{3}, 0, -\frac{1}{5}\right)$; (ii) (3 4 5 0); (ii) $\left(\frac{1}{4}, \frac{1}{2}, 0, \frac{1}{4}\right)$.

Ans. (i) not (since negative component); (ii) not (since do not add upto 1); (iii) yes

2. Find a scalar multiple of each vector, which is a probability vector.

   (i) $\left(2, \frac{1}{2}, 0, \frac{1}{4}, \frac{3}{4}, 0, 1\right)$; (ii) $\left(\frac{1}{3}, 2, \frac{1}{2}, 0, \frac{1}{4}, \frac{2}{3}\right)$; (iii) (1 2 3 4 5 6); (iv) $\left(\frac{1}{2}, \frac{2}{3}, 0, 2, \frac{5}{6}\right)$.

Ans. (i) $\frac{4}{18}$; (ii) $\frac{12}{45}$; (iii) $\frac{1}{21}$. Then required probability vectors are $\frac{4}{18}\left(2, \frac{1}{2}, 0, \frac{1}{4}, \frac{3}{4}, 0, 1\right) = \left(\frac{8}{18}, \frac{2}{18}, 0, \frac{1}{18}, \frac{3}{18}, 0, \frac{4}{18}\right)$;
(iv) multiply 6 : (3, 4, 0, 12, 5). Divide by $3 + 4 + 0 + 12 + 5 = 24$. Then probability vector $\left(\frac{3}{24} = \frac{1}{8}, \frac{1}{6}, 0, \frac{1}{2}, \frac{5}{54}\right)$.

3. Which matrices are stochastic

   (a) $\begin{pmatrix} \frac{1}{3} & \frac{2}{3} & \frac{4}{3} \\ \frac{1}{2} & 1 & \frac{1}{2} \end{pmatrix}$ (b) $\begin{pmatrix} \frac{15}{16} & \frac{1}{16} \\ \frac{2}{3} & \frac{4}{3} \end{pmatrix}$

   (c) $\begin{pmatrix} 1 & 0 \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$ (d) $\begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{4} & \frac{3}{4} \end{pmatrix}$

Ans. (a) not (square); (b) not adding to 1; (c) yes; (d) No (negative entry)

4. Which of the following matrices are regular

   (a) $A = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ 0 & 1 \end{pmatrix}$ (b) $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$;

   (c) $C = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 0 & 1 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}$ (d) $D = \begin{pmatrix} 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 0 & 1 & 0 \end{pmatrix}$

Ans. **(a)** Not regular, since 1 appears in the main diagonal.

(b) $B^2 = I$, $B^3 = B$, B is not regular, since 1 appears in the main diagonal

(c) c not regular, 1 appears on diagonal

(d) $D^2 = \begin{pmatrix} 0 & 1 & 0 \\ \frac{1}{8} & \frac{5}{16} & \frac{9}{16} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \end{pmatrix}$, $D^3 = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{5}{32} & \frac{41}{64} & \frac{13}{64} \\ \frac{1}{8} & \frac{5}{16} & \frac{9}{16} \end{pmatrix}$

D is regular since all the entries of D^3 are positive.

5. Find the unique fixed probability vector of each matrix

(a) $A = \begin{pmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{2}{5} & \frac{3}{5} \end{pmatrix}$ (b) $B = \begin{pmatrix} \frac{1}{4} & \frac{3}{4} \\ \frac{5}{6} & \frac{1}{6} \end{pmatrix}$

(c) $C = \begin{pmatrix} 0.2 & 0.8 \\ 0.5 & 0.5 \end{pmatrix}$

Ans. (a) $\left(\frac{6}{11}, \frac{5}{11}\right)$; (b) $\left(\frac{10}{19}, \frac{9}{19}\right)$; (c) $\left(\frac{5}{13}, \frac{8}{13}\right)$.

6. Find the unique fixed probability vector of

(a) $A = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 1 & 0 \end{pmatrix}$, (b) $B = \begin{pmatrix} 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \end{pmatrix}$

Ans. (a) $\left(\frac{2}{9}, \frac{6}{9}, \frac{1}{9}\right)$ (b) $\left(\frac{5}{15}, \frac{6}{15}, \frac{4}{15}\right)$

7. Given $P = \begin{pmatrix} 0 & 1 \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$ (a) find a unique fixed probability vector; (b) what matrix does P^n approach; (c) what vector does $\left(\frac{1}{4} \quad \frac{3}{4}\right) P^n$ approach.

Ans. (a) $\left(\frac{1}{3}, \frac{2}{3}\right)$; (b) $P^5 = \begin{pmatrix} 0.31 & 0.69 \\ 0.34 & 0.66 \end{pmatrix} \to$ (0.33, 0.66); (c) $\left(\frac{43}{128} \quad \frac{85}{128}\right) \approx \left(\frac{1}{3} \quad \frac{2}{3}\right)$

8. **(a)** Find the unique fixed probability vector t of

$$P = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{4} & 0 & \frac{1}{4} \\ 0 & 0 & 0 & 1 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix}$$

(b) What matrix does P^n approach

(c) What vector does $\left(\frac{1}{4}\ 0\ \frac{1}{2}\ \frac{1}{4}\right) P^n$ approach

(d) What vector does $\left(\frac{1}{2}\ 0\ 0\ \frac{1}{2}\right) P^n$ approach.

Hint: $t = (x, y, z, 1 - x - y - z)$, $y = 2x$, $x = 2z$, $y = 4z$

Ans. **(a)** $t = \left(\frac{2}{11}, \frac{4}{11}, \frac{1}{11}, \frac{4}{11}\right) = (0.1818, 0.3636, 0.0909, 0.3636)$

(b) $P^5 = \frac{1}{1024} \begin{pmatrix} 196 & 546 & 136 & 540 \\ 306 & 569 & 100 & 766 \\ 240 & 440 & 96 & 592 \\ 200 & 436 & 80 & 552 \end{pmatrix} \sim t$

(c) $\sim t$

(d) $\sim t$

9. Find the transition matrix

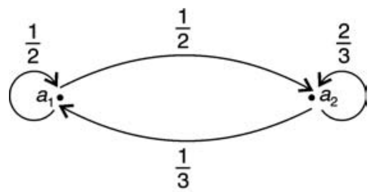

Fig. 31.10

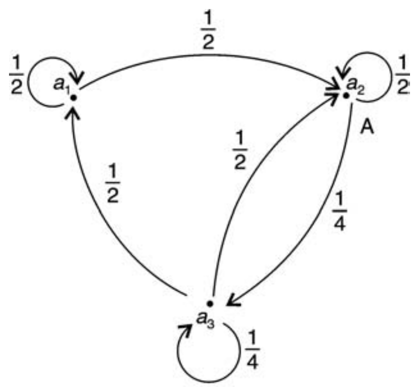

Fig. 31.11

Ans. **(i)** $\begin{array}{c|c} & \begin{matrix} a_1 & a_2 \end{matrix} \\ \begin{matrix} a_1 \\ a_2 \end{matrix} & \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix} \end{array}$

(ii) $\begin{array}{c|c} & \begin{matrix} a_1 & a_2 & a_3 \end{matrix} \\ \begin{matrix} a_1 \\ a_2 \\ a_3 \end{matrix} & \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \end{pmatrix} \end{array}$